
Course Title: C++ Programming II
Course Number: CSC275
Course Prerequisites: CSC215
Credit Hours: 3
General Studies Credits: 0



I. Course Description:

This course focuses on the craft of software. Students will build upon their previously acquired programming and language fundamentals and develop additional skills essential for crafting high quality and maintainable software. This is a project-based class and students will apply C++ and real world tools and practices to solve common programming problems involving advanced object oriented programming with design patterns, advanced Standard Template Libraries, and dynamic memory management including smart pointers, shared code, multithreading and network I/O.

II. Course Objectives:

Upon successful completion of this course, the student will have demonstrated competency in the following course learning objectives:

1. Declare variables using appropriate data-types & scope.
2. Write code to read and write text files
3. Write code to communicate over a network.
4. Import and utilize middleware libraries applications.
5. Write exception safe code.
6. Write multi-threaded code.
7. Write memory efficient code though smart pointers
8. Derive new classes from base classes and abstract classes.
9. Use polymorphism in applications.

III. Program Objectives:

The outcomes of this course are aligned and developed to support the following program learning objectives.

- **Advancing Computer Science**
 - Objectives 1: Follow a software development process to analyze a problem, and to design, build, and test software solutions.
 - Objectives 2: Demonstrate software development skills using more than one programming language and development environment.
 - Objectives 6: Within software solutions describe, implement and analyze data structure techniques such as lists, trees, hash tables, graphs, along with sorting and searching algorithms.
- **Artificial Intelligence**
 - Objective 5: Demonstrate software development skills using more than one programming language, development environment, platform, and source control system.
 - Objective 6: Describe, develop, analyze, integrate data structures, databases, and database management systems.
- **Data Science**
 - Objective 6: Within software solutions describe, implement and analyze data structure techniques such as lists, trees, hash tables, graphs, along with sorting and searching algorithms.
- **Game Programming**
 - Objective 4: Use software development processes to analyze a project problem, and to design, build, and test a corresponding software solution.
 - Objective 5: Demonstrate development skills using multiple programming languages, development environments, and platforms, including advanced and/or experimental topics in game programming.
- **Robotics and Embedded Systems**
 - Objective 1: Design and complete robotic and embedded systems solutions that apply to real-world situations and challenges.

IV. Boards Items:

The items listed below are identified as important and could be considered for boards presentation. These items demonstrate mastery of program learning objectives and are the same across all modalities of the course.

- Assignment: Multi-threading
- Assignment: Networking
- Final Project

V. Performance Evaluation:

Successful completion of this course requires students to obtain an overall cumulative grade of 60% or higher. Rating of the student's success in completing the stated course learning objectives will be based on the following percentiles:

Discussions:	20%
Assignments:	45%
Quizzes:	15%
Final Project:	20%

VI. Instructional Materials and References:

- Course shell
- Textbook
 - C++ Programming Fundamentals. ISBN: 978-1683929765

VII. Student Activity Expectation:

Students are awarded academic credit for this course by establishing expectations for in class, asynchronous, and out of class time.

On-Campus (15 Weeks)	Online (5 Weeks)	SyncFlex (7.5 Weeks)
Class Hours: • Term: 60 hours • Week: • In Class: 2 hours • Asynchronous: 2 hours Outside of Class: • Term: 90 hours • Week: 6 hours	Class Hours: • Term: 60 hours • Week: 12 hours Outside of Class: • Term: 90 hours • Week: 18 hours	Class Hours: • Term: 60 hours • Week: ◦ In Class: 4 hours ◦ Asynchronous: 4 hours Outside of Class: • Term: 90 hours • Week: 12 hours

VIII. Date of Last Review of Syllabus:

09/04/2025

The nature of course delivery necessitates some flex in the syllabus of the course. Courses can potentially vary from the stated syllabus but will cover all of the course learning objectives.